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1. Introduction

This report is about the first part of our group project. We are analyzing and planning a new computer system called the AI-Enabled Online Community Health Services Management System (AI-OCHSMS). Our main goal is to create a smart and easy-to-use digital system to improve health services in our community. We will use artificial intelligence (AI) to make the system work better, help people make decisions, and give them early health warnings.

Today, the government's health services are not digital. This causes problems with managing patient information, making appointments, and looking at health data. Our project will fix this by building one platform for everything. The AI-OCHSMS will not only put patient records on a computer but will also use smart features to help people communicate, guess health risks, and do simple jobs automatically. This will make public health services better for everyone.

The AI-OCHSMS is a management system that uses AI in important ways. It is not a tool for doctors to diagnose illnesses. The system will have different parts for citizens, healthcare workers, and people who manage the system. We chose three important AI features: an **AI Chatbot**, **Predictive Analytics**, and **Natural Language Processing (NLP)**. These AI features are important because they fix real problems and help the health services be more proactive and smarter.

2. Stakeholder Identification and System Objectives

2.1 Stakeholder Analysis

We've identified the main people and groups who will be involved with the AI-OCHSMS:

- **Citizens/Patients:** They are the end-users. They want an easy-to-use platform to book appointments, view their records, and get reliable health information.
- **Healthcare Workers (Doctors/Nurses):** They need an efficient and secure system to manage patient records, update vaccination statuses, and write prescriptions, which will save them time.
- **Health Officers/Admin Staff:** They are in charge of running the system. They need strong tools to manage users and services and to get useful reports and dashboards for planning.
- **Local Government:** They are the project's sponsor. They want to see that the system modernizes health services and follows all security and privacy laws.

2.2 System Objectives

Based on what the stakeholders need, our project's main goals are:

1. To create a secure and user-friendly platform that lets citizens manage their own health needs.
2. To improve the efficiency and accuracy of patient data management for healthcare workers.

3. To provide a powerful admin dashboard so that health officers can make smart, data-driven decisions.
4. To make community health services better by using AI to offer predictive insights and easy-to-access support.

3. Requirement Specification Document

The requirements for the AI-OCHSMS are specified using the **FURPS+** model, which categorizes them into functional and non-functional areas. This ensures all aspects of system quality and user needs are addressed.

3.1 Functional Requirements (F)

These requirements describe the core features and actions the system must perform.

- **User Management:** The system must allow users to sign up, log in, and manage their profiles.
- **Appointment Management:** The system must let citizens book, change, and cancel appointments. It must also send automatic reminders.
- **Patient Record Management:** The system must allow healthcare workers to securely manage and update patient records, including vaccination history and prescriptions.
- **Admin Tools:** The system must provide an admin panel for managing users, clinics, and services, and for generating health statistics reports.
- **AI Chatbot:** The system must provide a 24/7 AI-powered chatbot that can assist with symptom checking and general health information.
- **Predictive Analytics:** The system must analyze patient data to predict health risks and potential disease outbreaks.
- **Natural Language Processing (NLP):** The system must convert spoken notes from healthcare workers into written text for patient records.

3.2 Non-Functional Requirements (URPS+)

These requirements define the system's quality and characteristics.

- **Usability (U):** The user interface must be intuitive and easy to use on both computers and phones, requiring minimal training.
- **Reliability (R):** The system must maintain a high level of uptime (99.5%) to ensure continuous access to services.
- **Performance (P):** The system must be fast, with all data queries and page loads completing in under 2 seconds.
- **Supportability (S):** The system must be easy to maintain and update. It should have clear documentation for developers and administrators.
- **+(Plus):** This category includes other non-functional requirements not covered by FURPS.

- **Security:** All patient data must be encrypted both in transit and at rest. Access control will be role-based.
- **Scalability:** The system's design must handle a growing number of users and data without a loss in performance.
- **Data Integrity:** The system must have rules in place to ensure data is consistent and accurate.

3.3 AI-Specific Requirements

These are the requirements for the AI features:

- **AI Chatbot:** The system must include a 24/7 AI chatbot that can give health information and help with symptom checking.
- **Predictive Analytics:** The system must use AI to analyze patient data to predict health risks and potential disease outbreaks.
- **Natural Language Processing (NLP):** The system must have a feature that converts spoken notes from healthcare workers into written text for patient records.

3.4 Data for AI Components

The AI features need specific types of data to work correctly:

- **AI Chatbot:** Needs a structured database of health conditions, symptoms, and advice.
- **Predictive Analytics:** Needs access to anonymized historical and real-time patient data, like age, medical history, and location.
- **NLP:** Needs a large collection of medical terms and spoken dictations to train the model to transcribe accurately.

4. Feasibility Analysis (TELOS)

4.1 TELOS Analysis

We've looked at the project's viability from five different angles:

- **Technical:** This project is technically possible because we have access to modern tools and AI technologies.
- **Economic:** It's affordable for a school project because we can use free and open-source software for diagrams and mockups.
- **Legal:** We must be careful to follow data privacy laws since we're handling sensitive patient information. We will focus on strong security measures and getting user consent.
- **Operational:** The system will fit well into current health service operations and should make things more efficient.
- **Scheduling:** The 4-week timeline for Part 1 is realistic as long as we follow our work plan.

4.2 Ethical Considerations

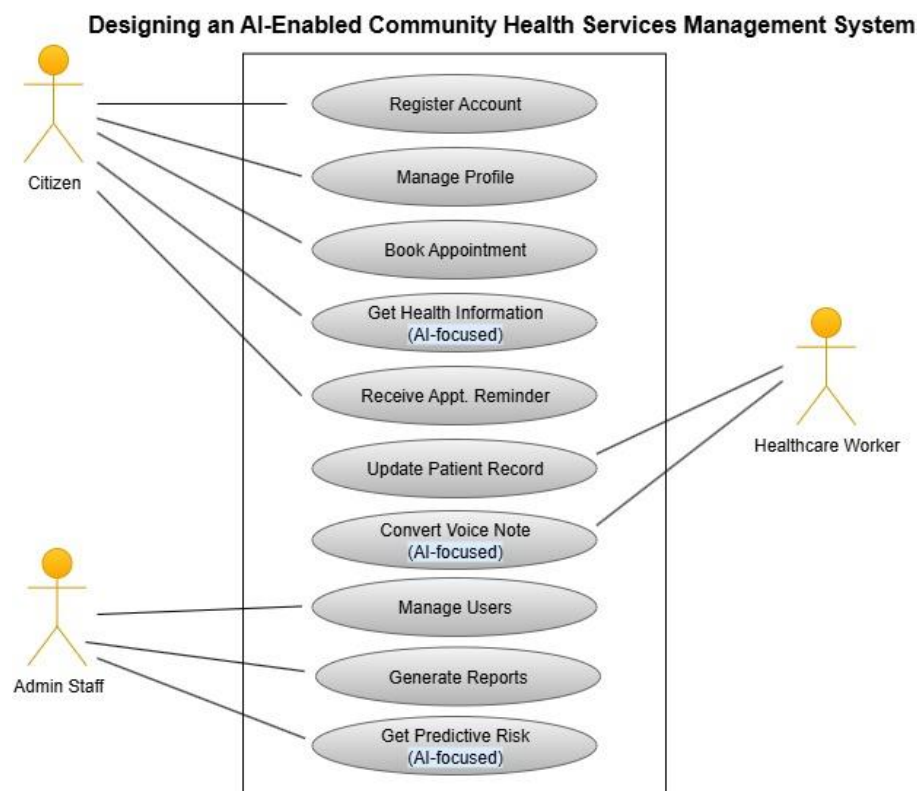
Using AI in healthcare brings up some important ethical points:

- **AI Bias:** The AI models must be trained on diverse data to make sure they are fair and don't favor certain groups of people.
- **Data Privacy and Security:** Patient data must be kept very safe. Our design will include strong security and a clear process for getting patient consent.
- **Transparency:** We need to be open with users about how the AI works. For example, if the AI gives a health alert, the system should explain why it did, without revealing personal data.

5. UML Diagrams

5.1 Use Case Diagram

This diagram shows the main users (Actors) and what they can do with the system (Use Cases). It includes our AI-focused use cases.



5.2 Use Case Descriptions

We'll provide a detailed written description for three of the main use cases:

Item	Description
Use Case ID	UC-01

Use Case Name	Book Appointment
Actors	Primary: Citizen; Secondary: System
Goal	To allow a registered citizen to book a health appointment using the system's AI recommendation feature.
Preconditions	Citizens must be logged into the system and have an active profile.
Postconditions	A new appointment record is successfully created, and the citizen receives a confirmation.
Trigger	Citizen clicks the "Book Appointment" button.
Main Flow	1. Citizen selects the service and preferred date. 2. System uses AI to analyze data and suggest optimal appointment times. 3. Citizen selects a time slot from the suggestions. 4. System displays a confirmation screen. 5. Citizen confirms the booking. 6. System creates a new appointment record and updates the database. 7. System sends an automated confirmation message.
Alternate Flows	AF-1: If the citizen cancels the booking, the system returns them to the main dashboard. AF-2: If no times are available, the system prompts the user to check other dates.
Exceptions	System error during booking, network connection is lost.
Special Requirements	The system must use AI to provide recommendations. Appointment confirmation must be sent in real-time.

Item	Description
Use Case ID	UC-02
Use Case Name	Convert Voice Note to Text
Actors	Primary: Healthcare Worker; Secondary: System
Goal	To allow a healthcare worker to dictate and transcribe notes directly into a patient's record.
Preconditions	Healthcare workers must be logged in and have a patient's record open.
Postconditions	The patient's record is updated with a new, structured clinical note.
Trigger	Healthcare workers select the "Start Voice Note" option.
Main Flow	1. Worker selects "Start Voice Note." 2. System activates the microphone and begins capturing audio. 3. Worker dictates clinical notes. 4. System's NLP component processes and transcribes the audio in real-time. 5. The transcribed text appears in a text field for review. 6. Worker reviews and makes any necessary edits. 7. Worker saves the note. 8. System updates the patient's digital record.
Alternate Flows	AF-1: If the worker stops recording before saving, the system discards the

	note.
Exceptions	Microphone fails; system is unable to transcribe due to poor audio quality.
Special Requirements	The transcription must be processed in real-time. The system must handle specialized medical terminology accurately.

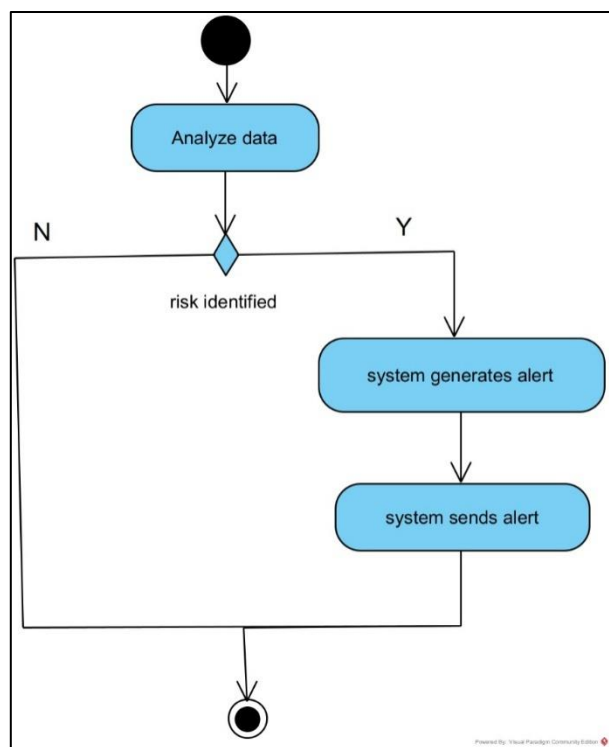
Item	Description
Use Case ID	UC-03
Use Case Name	Get Predictive Risk
Actors	Primary: Admin Staff; Secondary: System
Goal	To allow the system to use AI to analyze data and provide risk alerts to the admin staff for proactive decision-making.
Preconditions	A scheduled background task is initiated. The system has access to anonymized patient data.
Postconditions	The Admin Staff's dashboard is updated with a new risk alert.
Trigger	A pre-scheduled time for data analysis is reached.
Main Flow	1. A scheduled background task begins. 2. The AI model analyzes historical and real-time patient data. 3. The model identifies a risk pattern (e.g., a potential outbreak). 4. The system generates a risk alert. 5. System sends the alert to the Admin Staff dashboard. 6. Admin Staff view the alert and can take action.
Alternate Flows	AF-1: If no risk pattern is identified, the process ends without generating an alert.
Exceptions	System database or network error prevents the analysis from completing.
Special Requirements	Analysis must be based on a diverse dataset to avoid bias. The system must not store personal health information with the alert.

6. Activity Diagrams

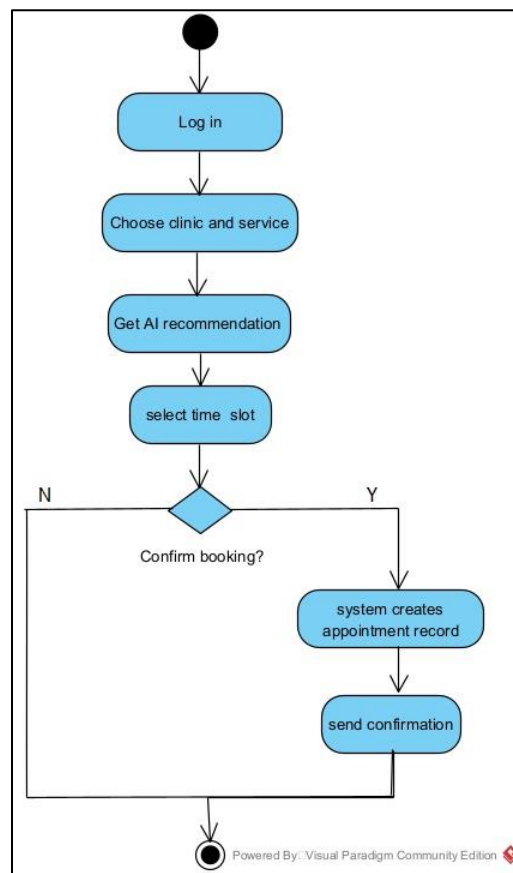
These diagrams will show the step-by-step process for key activities. We'll have one for:

- **Book Appointment:** Illustrates the citizen's flow from logging in to getting a confirmation.
- **Patient Data Update:** Shows the healthcare worker's flow for updating a record, including the option to use the NLP feature.
- **Predictive Analytics Alert:** Shows the automated process of the AI analyzing data and sending an alert to the admin dashboard.

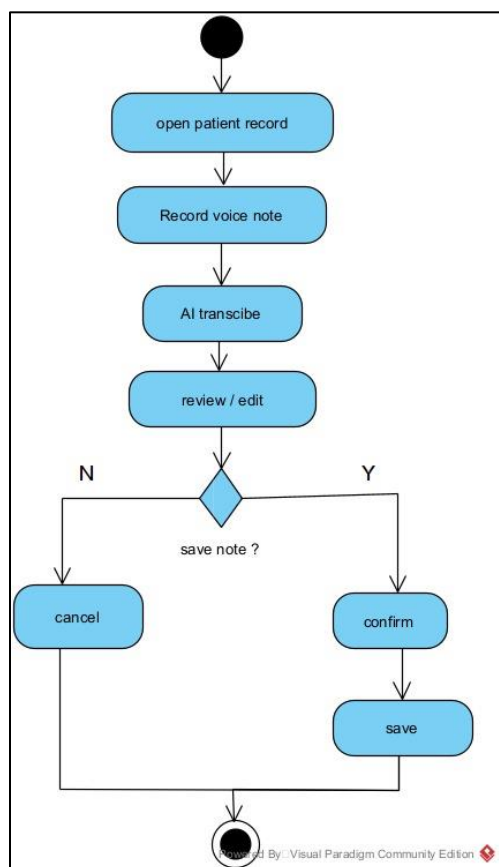
Predictive Analytics Alert



Book Appointment



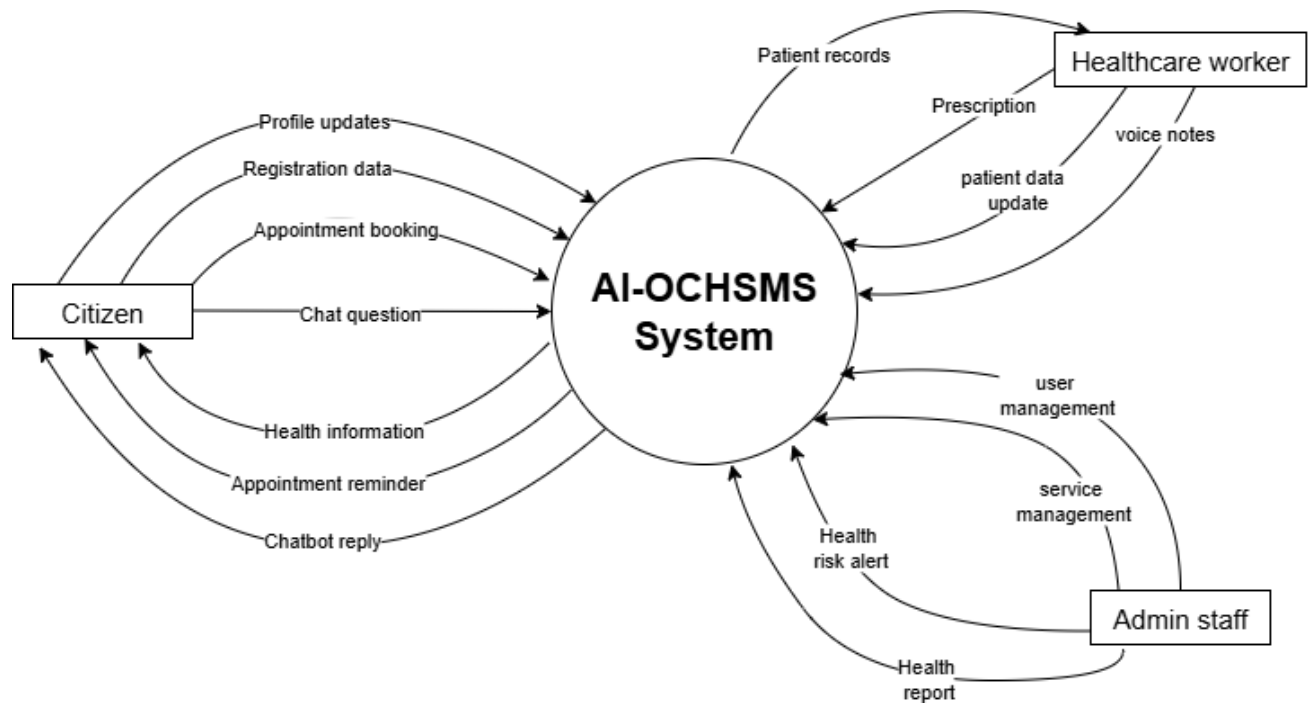
Patient Data Update



7. Data Flow Diagrams (DFDs)

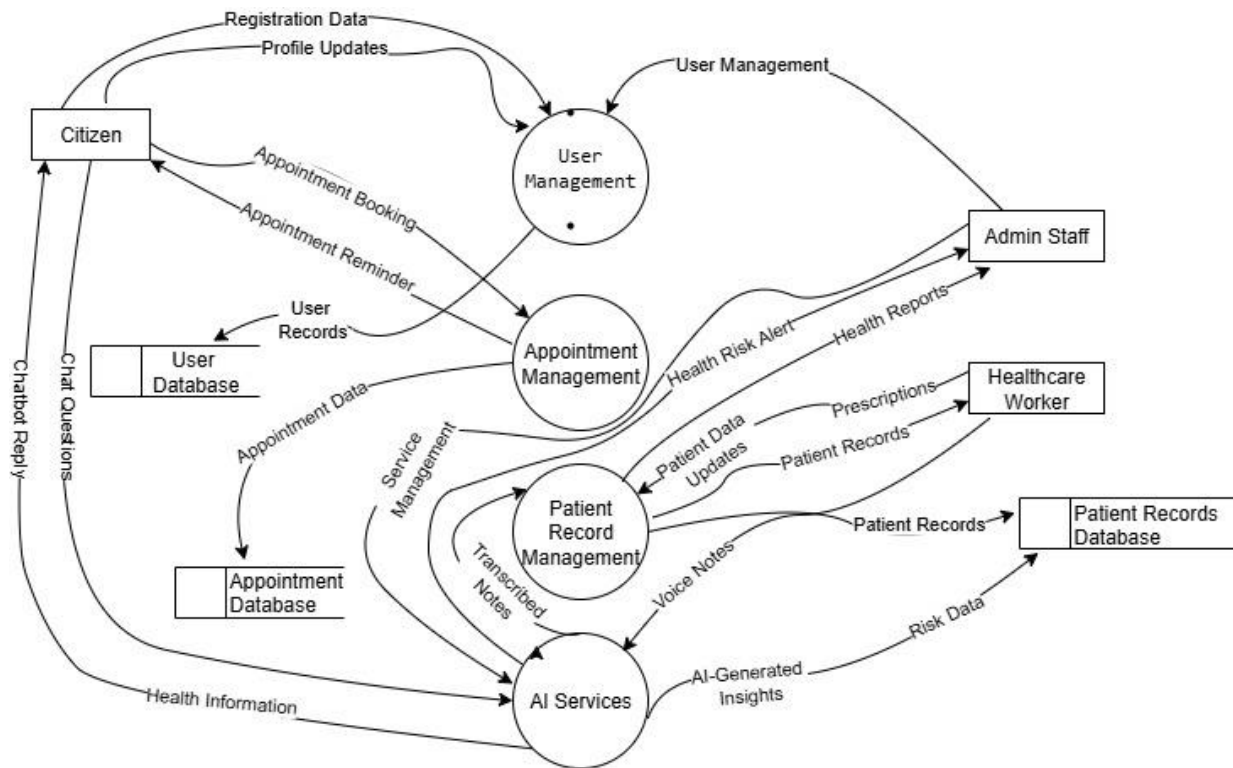
7.1 Context Diagram

This is a high-level diagram showing the entire system as one box. It shows the main users and the flow of information to and from the system.



7.2 Level 1 DFD

This diagram breaks down the main system into its sub-processes. It will show how data moves between these processes, our databases, and the users.



8. High-level System Scope and Limitations

- This section defines the boundaries of the AI-OCHSMS, explaining what the system is designed to do and what it will not do.
- **Scope:** The system's scope is focused on improving community health management. It will include core functionalities for citizens (appointment booking, profile management), healthcare workers (patient record updates), and administrators (user management, reports). The system's key feature is the integration of AI for automation and insights.
- **Limitations:** The AI-OCHSMS is designed as a management and information platform, not as a diagnostic tool. The AI Chatbot will provide general health information and symptom checking, but it will not give medical diagnoses or personalized advice. Furthermore, the system is designed for community health services, so it may not handle all the complex needs of a large hospital. The predictive analytics feature will provide alerts for potential risks, but it is not a guarantee of a healthy outcome.